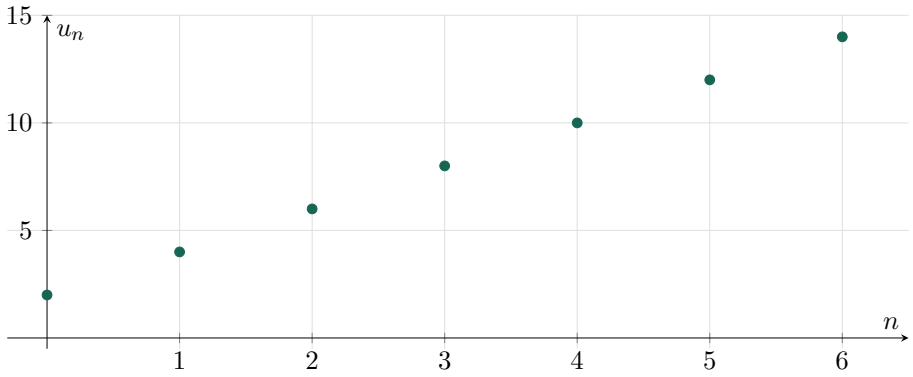
1.1 Defining and Representing a Succession

Definition and results

A real sequence is a function whose variable is an integer: at each row n authorized corresponds to a real u_n. Rating (u_n) means the continuation, u_n a term. An explicit formula gives directly u_n according to the n. A definition by induction provides a first term and a rule of passage. Representation is a set of points ($n; u_n$) : one does not connect necessarily these points.

Status and justification. Definition. The uniqueness of terms obtained from a first term and a deterministic rule is accepted; no formal induction reasoning is required.

Example solved. For $u_n = n^2 + 1$, $u_0 = 1$, u_3 = 10. For v_0 = 1 and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, v_3 = 15. We can't replace v_3 by 2 $\times$ 3 + 1 : the rule relates to the previous term.



Terms of $u_n = 2n$ + 2 : only the integer rows are represented.

Point of vigilance

The row is not the value. A finished list of terms does not determine a single rule.

Immediate application. Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; propose an explicit formula.

1.2 Arithmetic Sequences

Definition and results

A sequence is arithmetical of common difference r if $u_{n+1} = u_n + r$ for any authorized rank. $u_n = u_p + (n-p)r$ for $n \geq p$. If $r > 0$, it is strictly increasing; if $r < 0$, strictly Decreasing; if r = 0, constant. Points are aligned on a right.

Status and justification. For p fixed, the continuation $v_n = u_p + (n-p)r$ checks $v_p = u_p$ and $v_{n+1} = v_n + r$. By the uniqueness of successive terms, $u_n = v_n$. Variation is the result of the $u_{n+1} - u_n = r$.

Example solved. A reserve contains 140 L and receives 12 L per minute. n minutes, $u_n = 140+12n$. At 5 min: 200 L. If we start indexing at 1 with 140 L, we write $u_n = 140+12(n-1)$.

Point of vigilance

Adding a constant percentage does not usually give an arithmetic sequence.

Immediate application. We know. u_3 = 17 and u_8 = 37 for an arithmetic sequence. r and u_0.

1.3 Geometric sequences

Definition and results

A sequence is geometrical of common ratio q if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$with the agreement that an empty product is worth 1. A constant rate t gives $q = 1 + t$. Variation decides with $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

Status and justification. The formula gives the first term and its multiplication by q gives the next term. The uniqueness of the generation justifies the formula. q = 0, all terms after first are null; deal with this case directly.

Example solved. A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. Terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We get u_2 = 90 mg, not to be confused 25% decrease and 25 mg withdrawal.

Point of vigilance

$q > 1$ is not enough to conclude that any geometric sequence is increasing: the sign of the first term counts.

Immediate application. Studying the variation in $u_n = -2 \times 3^n$and then $v_n = 2(-1)^n$.

1.4 Amounts, indices and demonstrations

Definition and results

For $n \geq 1$, $1+2+\cdots+n = \frac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1+q+\cdots+q^n = \frac{1-q^{n+1}}{1-q}$. For $q = 1$, the sum is $n+1$. Thus, u_p to u_n, an arithmetic sum is $\frac{(n-p+1)(u_p+u_n)}{2}$; a geometric sum is $\frac{u_p(1-q^{n-p+1})}{1-q}$ if $q \neq 1$.

Status and justification. Writing $S = 1 + \cdots + n$ in both directions, each pair of the sum 2 S value $n + 1$, with n pairs. $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$: $(1-q)T = 1 - q^{n+1}$. The case $q = 1$ must precede division.

Example solved. For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \frac{5(7+15)}{2}$ = 55. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \frac{2(1-3^4)}{1-3} = 80$.

Point of vigilance

Count the terms before using the formula: from 0 to n, there are n+1.

Immediate application. Calculate $\sum_{k=2}^{5} 2^k$ without summation symbol in the answer.

1.5 Conjecture a limit and schedule a threshold

Definition and results

A limit is discussed here intuitively. $\frac{1}{n+1}$ close to 0; n^2 Grows without limit; $(-1)^n$ a digital experience suggests a conclusion but does not prove behavior for all ranks. A threshold algorithm must specify the a sense of inequality and provision for a stop.

Status and justification. Illustrative findings, without formal theory of limits. integer threshold is verified by checking the row found and the previous row, when monotonicity is established.

Example solved. For $u_n = 80(\frac{1}{2})^n$, the first ranking such as $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The programme begins with $n = 0, u$ = 80 and repeats division by 2 as long as $u \geq 6$.

Point of vigilance

A terminalless loop may never stop if the threshold is inaccessible; $<$ and $\leq$ are not Interchangeable.

Immediate application. For $u_n = 48(\frac{1}{2})^n$, determine the first rows where $u_n \leq 6$ and where $u_n < 6$.